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Logical Functions

The following logical functions evaluate an expression and return a specific result.

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AND Function

The AND function evaluates logical conditions. If all of its conditions are TRUE, the function will return TRUE. If 1 or more of its conditions is FALSE, the function will return FALSE. The AND function must be used in conjunction with an IF function.

Return Type: TRUE or FALSE

Syntax: AND(**logical1**, **logical2**,...)

In the above syntax, parameters in bold are required.

The following table describes AND function parameters.

Parameter	Description
logical1, logical2, and so on	Conditions that can be evaluated to TRUE or FALSE. This condition can be written with any comparison operator (=, <, >, <=, >=, <>). An example of how this parameter might be formatted is [Field 1]>20.

Examples:

The following table provides example formulas of the AND function.

Formula	Result
IF(AND([Age] > 1, [Age] < 50)) where the value in the Age field is 35.	TRUE
IF(AND([Severity] = 10, [Impact] > 7)) where the value in the Severity field is 10 and the value in the Impact field is 3.	FALSE

IF Function

The IF function evaluates a logical condition, and if the condition is TRUE, 1 value is returned. If the condition is FALSE, another value is returned. The IF function can also be nested to construct more elaborate tests, as shown in the following example:

IF([Rating]>15,"A", IF([Rating]>10,"B", IF([Rating]>5," C")))

For more information on nesting IF functions, see the fourth example in the "Examples" section below.

Return Type: Text, numeric, date or a Values List field selection, depending on the type of data supplied for the value_if_true and value_if_false parameters

Syntax: IF(**logical_test**, **value_if_true**, value_if_false)

In the above syntax, parameters in bold are required.

The following table describes IF function parameters.

Parameter	Description
logical_test	Any condition that can be evaluated to TRUE or FALSE. This condition can be written with any comparison operator (=, <, >, <=, >=, <>). An example of how this parameter might be formatted is [Field 1]>20.
value_if_true	The value that is returned if the logical_test parameter is TRUE. This parameter can be formatted as a text string, such as "High Risk", or as a Values List field selection, such as VALUEOF("Urgent"). The parameter can also be formatted as another formula, such as SUM([Field 1],[Field 2]).
value_if_false	The value that is returned if the logical_test parameter is FALSE. This parameter can be formatted as a text string, such as "Low Risk", or as a Values List field selection, such as VALUEOF("Not Urgent"). The parameter can also be formatted as another formula, such as SUM([Field 1],[Field 3]). If this parameter is omitted from the formula and the logical_test parameter evaluates to FALSE, the calculated field will return empty (no value).

Examples:

The following table provides example formulas of the IF function.

Formula	Result
IF([Days Since Last Virus Scan] > 1, "High risk", "Low risk") where the value in the Days Since Last Virus Scan field is 3.	High risk
IF([Rating] = 10, "Follow up") where the value in the Rating field is 7.	The field will return empty.
IF([Severity] >= 10, VALUEOF("Urgent"), VALUEOF("Not Urgent")) where the value in the Severity field is 10.	Urgent
IF([Rating] > 15, "A", IF([Rating] > 10, "B", IF([Rating] > 5, "C"))) where the value in the Rating field is 12. Note: In this example of nested IF statements, the second IF statement serves as the value_if_false parameter to the first IF statement, and the third IF statement serves as the value_if_false parameter to the second IF statement. Because the value of the Rating field in this example is 12, the first IF statement does not prove TRUE, so the second IF statement is evaluated and, in this case, proves TRUE. If the value of the Rating field were 8, the second IF statement would also prove FALSE, and the third IF statement would be evaluated.	B

NOT Function

The NOT function evaluates a logical condition. If the condition is TRUE, the function returns the value of FALSE. If the condition is FALSE, the function returns the value of TRUE. Use the NOT function when you want to ensure that a value is not equal to 1 particular value.

Return Type: TRUE or FALSE

Syntax: NOT(logical_test)

In the above syntax, parameters in bold are required.

The following table describes the NOT function parameter.

Parameter	Description
logical_test	Any condition that can be evaluated to TRUE or FALSE. This condition can be written with any comparison operator (=, <, >, <=, >=, <>). An example of how this parameter might be formatted is [Field 1]>20.

Examples:

The following table provides example formulas of the NOT function.

Formula	Result
NOT([Rating] = 10) where the value in the Rating field is 10.	FALSE
NOT([Number of Clients in Attendance] > 20) where the value in the Number of Clients in Attendance field is 12.	TRUE

OR Function

The OR function evaluates logical conditions. If any of the condition evaluates to TRUE, the function returns the value of TRUE. If none of conditions evaluate to TRUE, the function returns the value of FALSE.

Return Type: TRUE or FALSE

Syntax: OR(logical1, logical2,...)

In the above syntax, parameters in bold are required.

The following table describes OR function parameters.

Parameter	Description
logical1, logical2, and so on	Conditions that can be evaluated to TRUE or FALSE. This condition can be written with any comparison operator (=, <, >, <=, >=, <>). An example of how this parameter might be formatted is [Field 1]>20.

Examples:

The following table provides example formulas of the OR function.

Formula	Result
OR([Risk] = 4, [Criticality] = 7) where the value in the Risk field is 4 and the value in the Criticality field is 2.	True (because 1 of the 2 parameters evaluated TRUE)
OR([Risk] = 4, [Criticality] = 7) where the value in the Risk field is 9 and the value in the Criticality field is 5.	False (because both of the parameters evaluated FALSE)

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